

Md Nafieu Hossain Alif

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EDUCATION

McMaster University

Bachelor of Software Engineering CO-OP | GPA: 3.8/4.0

Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Architecture, Operating Systems

Hamilton, ON

Expected May 2027

EXPERIENCE

Software Engineer Intern

NAVBLUE (Airbus Company)

May 2024 – Present

Waterloo, ON

- Optimized critical flight operations APIs serving 50,000+ daily requests, reducing average response time by 45% (2.3s → 1.2s) through **SQL** optimization and **Redis** caching, impacting 200+ airline customers.
- Architected database migration strategy for 3 production environments handling 2M+ flight records, eliminating 6 hours of weekly downtime via automated rollback procedures.
- Built comprehensive testing framework with 95%+ code coverage using **xUnit** and **Python Behave**, cutting production bugs by 60% and enabling a **CI/CD pipeline** that reduced release cycle from 2 weeks to 3 days.
- Led crisis response for a Tier-1 customer experiencing 40% performance degradation; identified **AWS RDS** connection pooling issue and delivered a fix within 8 hours, preventing \$500K+ revenue impact.
- Designed microservice integration layer connecting 12 internal services to the **Flight Ops Event Bus**, reducing inter-service latency by 30% and enabling real-time sync for 100,000+ flight operations.

Web Development Lead

Google Developer Students Club McMaster

Sep. 2023 – Present

Hamilton, ON

- Spearheaded full-stack development of the GDSC portal and hackathon platform (**MERN**), growing active users from 150 to 800+ students (430% increase) in 6 months.
- Implemented performance optimizations reducing page load times by 55% (4.2s → 1.9s), yielding 40% higher session duration and 25% more weekly active users (**Google Analytics**).
- Launched a gamified learning platform with a 5-person team, achieving 300+ positive reviews and 85% retention in the first month; adopted by 3 other GDSC chapters.
- Technical interviewer: screened **30+** candidates with **80%** hire accuracy. Mentored in **Node/React** best practices.

PROJECTS

AegisMark — Invisible Watermarking System | *FastAPI, TensorFlow, GCP, Kubernetes*

Aug 2025

- Built a cloud-native watermarking system preserving 99.2% **SSIM** and 44+ dB **PSNR** across 10,000+ test images.
- Deployed scalable **FastAPI** service on **GCP Kubernetes**, securing digital ownership via cryptographic watermarks resilient to edits and compression.
- Trained **encoder-decoder CNNs** with **ECC** to embed 128-bit payloads and achieve sub-30 ms inference.
- Integrated automated robustness tests for cropping, and compression, maintaining **98%** watermark recoverability.
- Implemented vectorized inference pipeline with **TensorFlow Serving**, boosting throughput by 2.1× under load.

ChatPBS — RAG Flight Operations Assistant | *FastAPI, LangChain, FAISS, PostgreSQL*

June 2024

- Developed a RAG-based chatbot enabling 20+ pilots to query 1,000+ proprietary documents with cited answers.
- Boosted retrieval accuracy from 42%→83% via **LangChain** orchestration and hybrid search using **FAISS**.
- Built ingestion pipeline for semantic PDF chunking and vector indexing, reducing lookup latency to **1.3s**.
- Deployed containerized **FastAPI** services on **AWS ECS** with **Redis** caching, sustaining 99.5% uptime.
- Won **1st place** at NAVBLUE Spring Hackathon and initiated roadmap integration discussions for 2025.

TECHNICAL SKILLS

Languages: C#, Python, Java, JavaScript/TypeScript, SQL, C, C++, Ruby, Go, Verilog, Assembly

Frameworks: .NET Core, Next.js, Flask, Spring Boot, GraphQL, TensorFlow, PyTorch, FastAPI, Apache Kafka

ML/AI: PPO, DQN, LlamaIndex, Huggingface, MLflow, SageMaker, OpenAI Gym, LangChain, LangGraph

Cloud: AWS, Azure, Docker, Kubernetes, GCP, RunPod, SQS, S3, Cloudformation, Terraform, Grafana

Databases: MongoDB, MySQL, Oracle, NoSQL, Microsoft SQL Server, PostgreSQL, Redis

Tools: Gitlab CI, Jira, VS Code Tools, JetBrains Tools, Insomnia, Confluence